DataScience Mini Project - Solar Sense

Predicting Solar Energy Generation Using Weather Data

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1. Introduction

Solar energy generation is highly dependent on environmental and weather parameters. This project aims to predict the AC Power output (Solar Energy Generation) using various weather features. By successfully forecasting the energy generated, grid operators can optimize energy distribution and maintain balance.

2. Data Collection and Loading

Two primary datasets were utilized for this investigation:

- 1. Plant_1_Generation_Data.csv: Contains metrics related to the power output of the solar plant.
- 2. Plant_1_Weather_Sensor_Data.csv: Contains weather-related parameters such as ambient temperature and irradiation levels.

These datasets were merged on the 'DATE_TIME' column as connecting weather data to generating data is crucial for analyzing dependencies.

3. Data Preprocessing

The merged dataset was subjected to comprehensive preprocessing techniques:

Missing Values Handling: Forward-fill techniques were implemented to fill any gaps in the time-series data without losing contextual timelines.

Removing Duplicates: Identical records were pruned to ensure unbiased model weights.

Feature Engineering: The 'DATE_TIME' field was decomposed into distinct numerical components—'HOUR', 'DAY', and 'MONTH'. These represent key cyclical features influencing solar intensity.

Column Pruning: Unnecessary identifier columns (e.g., 'SOURCE_KEY') were dropped to reduce model complexity.

4. Exploratory Data Analysis (EDA)

Extensive visual exploration was conducted to identify underlying patterns between dependent and independent variables:

- Correlation Heatmap: Demonstrated the strong positive relationship between irradiation levels and solar power output.
- Time-Series Analysis: Displayed solar power fluctuations throughout the day, aligning closely with standard daylight cycles.

- Scatter Plots: Highlighted distinct clusters representing how Ambient Temperature and Irradiation affect power generation. Irradiation proved to be the most influential predictor.

5. External Data Visualization

The cleaned dataset was exported ('cleaned_solar_data.csv') to advanced Business Intelligence (BI) tools for creating comprehensive analytical dashboards. Features such as IRRADIATION were transformed (e.g., scaled by 1000) for better interpretability in dashboards.

6. Model Development

The target of our system is to predict the 'AC_POWER'. The predictive model was architected via the following steps:

Feature Selection: 'IRRADIATION', 'AMBIENT_TEMPERATURE', and 'HOUR' were selected as independent variables (X), with 'AC_POWER' as the dependent target (y).

Train-Test Split: An 80-20 split strategy was implemented using sklearn's train_test_split algorithm.

Random Forest Regressor: Chosen for its robustness against overfitting and excellence in capturing non-linear relationships. The model was trained on the training partition and validated against the unseen test data.

7. Results and Conclusion

The Random Forest Regressor's performance was evaluated using standard regression metrics:

- R^2 (R-squared) Score: Indicated a strong fit of the model logic to actual power generation constraints.
- MAE (Mean Absolute Error): Measured the average magnitude of absolute errors in predictions.

Conclusion:

The predictive model effectively utilizes weather data—specifically irradiation and temperature—to accurately estimate solar plant outputs. This tool can heavily aid energy sectors in anticipating grid supplies. Future enhancements could explore more advanced models such as Gradient Boosting or time-series explicit models like LSTMs to capture temporal dependencies.